O*NET OnLine

Database Administrators

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Administer, test, and implement computer databases, applying knowledge of database management systems. Coordinate changes to computer databases. Identify, investigate, and resolve database performance issues, database capacity, and database scalability. May plan, coordinate, and implement security measures to safeguard computer databases.

Sample of reported job titles: Database Administration Manager, Database Administrator (DBA), Database Analyst, Database Coordinator, Database Engineer, Database Manager, Development and Database Administration Manager, IS Admin (Information Systems Administrator), Systems Administrator (Systems Admin)

Occupation-Specific Information

Tasks

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- Modify existing databases and database management systems or direct programmers and analysts to make changes.
- Plan, coordinate, and implement security measures to safeguard information in computer files against accidental or unauthorized damage, modification or disclosure.
- Plan and install upgrades of database management system software to enhance database performance.
- Specify users and user access levels for each segment of database.
- Test changes to database applications or systems.

Occupational Requirements

Work Activities

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- **Working with Computers** — Using computers and computer systems (including hardware and software) to program, write software, set up functions, enter data, or process information.

- **Making Decisions and Solving Problems** — Analyzing information and evaluating results to choose the best solution and solve problems.
- **Getting Information** — Observing, receiving, and otherwise obtaining information from all relevant sources.
- **Identifying Objects, Actions, and Events** — Identifying information by categorizing, estimating, recognizing differences or similarities, and detecting changes in circumstances or events.
- **Analyzing Data or Information** — Identifying the underlying principles, reasons, or facts of information by breaking down information or data into separate parts.

Detailed Work Activities

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- Create databases to store electronic data.
- Update computer database information.
- Implement security measures for computer or information systems.
- Develop computer or information security policies or procedures.
- Install computer software.

Work Context

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- **E-Mail** — 92% responded “Every day.”
- **Indoors, Environmentally Controlled** — 90% responded “Every day.”
- **Telephone Conversations**
- **Work With or Contribute to a Work Group or Team**
- **Freedom to Make Decisions** — 56% responded “A lot of freedom.”

Experience Requirements

Job Zone

Title	Job Zone Four: Considerable Preparation Needed
Education	Most of these occupations require a four-year bachelor's degree, but some do not.
Related Experience	A considerable amount of work-related skill, knowledge, or experience is needed for these occupations. For example, an accountant must complete four years of college and work for several years in accounting to be considered qualified.
Job Training	Employees in these occupations usually need several years of work-related experience, on-the-job training, and/or vocational training.
Job Zone Examples	Many of these occupations involve coordinating, supervising, managing, or training others. Examples include real estate brokers, sales managers, database

administrators, graphic designers, conservation scientists, art directors, and cost estimators.

SVP Range (7.0 to < 8.0)

Training & Credentials

State training Select a State Go

Local training ZIP Code Go

Certifications Find certifications nationwide

Apprenticeship Opportunities

All 2 displayed

Example apprenticeship titles for this occupation:

- Database Administrator (Nof)
- Database Technician

Specific title(s) listed above are vetted by industry and approved by the U.S. Department of Labor for use in a Registered Apprenticeship Program.

Start your career and build your skillset. Visit **Apprenticeship.gov** to learn about opportunities related to this occupation.

Worker Requirements

Software Skills

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- **Data base management system software** — Amazon DynamoDB; Elasticsearch; MongoDB; Oracle PL/SQL
- **Data base user interface and query software** — Apache Hive; Blackboard software; IBM DB2; MySQL
- **Development environment software** — Apache Kafka; Apache Maven; Go; Oracle SQL Developer
- **Object or component oriented development software** — C#; jQuery; Scala; Swift
- **Web platform development software** — Django; Google Angular; Microsoft ASP.NET; Spring Framework

Hot Technologies are requirements most frequently included across all employer job postings.
See all 112 Hot Technologies for this occupation.

In Demand skills are frequently included in employer job postings for this occupation.
See all 38 In Demand skills for this occupation.

Essential Skills

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- **Critical Thinking** — Using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions, or approaches to problems.
- **Active Listening** — Giving full attention to what other people are saying, taking time to understand the points being made, asking questions as appropriate, and not interrupting at inappropriate times.
- **Reading Comprehension** — Understanding written sentences and paragraphs in work-related documents.
- **Active Learning** — Understanding the implications of new information for both current and future problem-solving and decision-making.
- **Speaking** — Talking to others to convey information effectively.

Transferable Skills

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- **Complex Problem Solving** — Identifying complex problems and reviewing related information to develop and evaluate options and implement solutions.
- **Judgment and Decision Making** — Considering the relative costs and benefits of potential actions to choose the most appropriate one.
- **Programming** — Writing computer programs for various purposes.
- **Operations Analysis** — Analyzing needs and product requirements to create a design.
- **Systems Analysis** — Determining how a system should work and how changes in conditions, operations, and the environment will affect outcomes.

Knowledge

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- **Computers and Electronics** — Knowledge of circuit boards, processors, chips, electronic equipment, and computer hardware and software, including applications and programming.
- **English Language** — Knowledge of the structure and content of the English language including the meaning and spelling of words, and rules of composition and grammar.
- **Customer and Personal Service** — Knowledge of principles and processes for providing customer and personal services. This includes customer needs assessment, meeting quality standards for services, and evaluation of customer satisfaction.
- **Mathematics** — Knowledge of arithmetic, algebra, geometry, calculus, statistics, and their applications.
- **Telecommunications** — Knowledge of transmission, broadcasting, switching, control, and operation of telecommunications systems.

Education

How much education does a new hire need to perform a job in this occupation? Respondents said:

89% Bachelor's degree required

4% Post-baccalaureate certificate required

3% Associate's degree or other 2-year degree required

Worker Characteristics

Abilities

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- **Deductive Reasoning** — The ability to apply general rules to specific problems to produce answers that make sense.
- **Inductive Reasoning** — The ability to combine pieces of information to form general rules or conclusions (includes finding a relationship among seemingly unrelated events).
- **Information Ordering** — The ability to arrange things or actions in a certain order or pattern according to a specific rule or set of rules (e.g., patterns of numbers, letters, words, pictures, mathematical operations).
- **Oral Comprehension** — The ability to listen to and understand information and ideas presented through spoken words and sentences.
- **Problem Sensitivity** — The ability to tell when something is wrong or is likely to go wrong. It does not involve solving the problem, only recognizing that there is a problem.

Interests

All 2 displayed

Interest code: **CI**

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- **Conventional** — Work involves following procedures and regulations to organize information or data, typically in a business setting. Conventional occupations are often associated with office work, accounting, mathematics/statistics, information technology, finance, or human resources.
  - **Information Technology** — Work involves designing, developing, testing, and maintaining computer software, hardware, networks, and systems.
  - **Mathematics/Statistics** — Work involves developing and applying mathematics and statistical theory and methods.
- **Investigative** — Work involves studying and researching non-living objects, living organisms, disease or other forms of impairment, or human behavior. Investigative occupations are often associated with physical, life, medical, or social sciences, and can be found in the fields of humanities, mathematics/statistics, information technology, or health care service.

- **Mathematics/Statistics** — Work involves developing and applying mathematics and statistical theory and methods.

Work Styles

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- **Intellectual Curiosity** — A tendency to seek out and acquire new work-related knowledge and obtain a deep understanding of work-related subjects.
- **Cautiousness** — A tendency to be careful, deliberate, and risk-avoidant when making work-related decisions or doing work.
- **Integrity** — A tendency to be honest and ethical at work.
- **Attention to Detail** — A tendency to be detail-oriented, organized, and thorough in completing work.
- **Dependability** — A tendency to be reliable, responsible, and consistent in meeting work-related obligations.

Workforce Characteristics

Wages & Employment Trends

Median wages (2025)	$50.30 hourly, $104,620 annual
State wages	Select a State Go
Local wages	ZIP Code Go
Employment (2025)	75,000 employees
Projected growth (2025-2035)	Little or no change
Projected job openings (2025-2035)	3,300
State trends	Select a State Go
Top industries (2025)	Professional, Scientific, and Technical Services Finance and Insurance

Source: Bureau of Labor Statistics 2025 wage data and 2025-2035 employment projections. "Projected growth" represents the estimated change in total employment over the projections period (2025-2035). "Projected job openings" represent openings due to growth and replacement.

Job Openings on the Web

State job openings Select a State Go

Local job openings ZIP Code Go

More Information

Related Occupations

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- 15-1211.00 Computer Systems Analysts Bright Outlook
- 15-1299.08 Computer Systems Engineers/Architects
- 15-1243.00 Database Architects
- 15-1244.00 Network and Computer Systems Administrators
- 15-1252.00 Software Developers

Professional Associations

Disclaimer: Sources are listed to provide additional information on related jobs, specialties, and/or industries. Links to non-DOL Internet sites are provided for your convenience and do not constitute an endorsement.

National Associations

- Association for Computing Machinery
- Computing Research Association
- IEEE Computer Society
- National Center for Women and Information Technology

Accreditation, Certification, & Unions

- CompTIA
- Institute for Certification of Computing Professionals